CHAPTER 2

LITERATURE REVIEW

Bhandari (2016) defines a convention center as any property utilized for an event concerning a conference, meeting center, or any similar facility; this includes administrative offices, exposition halls, auditoriums, and facilities for preparing and serving food. Sung & Lee (2015), on the other hand, defines a convention center as a sizable building that is designed to hold conventions, exhibitions, trade shows, as well as other events.

Because the conference industry is comparatively young, its terminology remains loosely defined; the term 'convention center' is sometimes used interchangeably with 'exhibition center' and related terms such as congress, assembly, or symposium, yet these venues share a common purpose: bringing people together for the exchange of ideas and socialization (Rogers & Wynn-Moylan, 2022; Sharma, 2025).

Convention centers play a significant part in the MICE industry (Meetings, Incentives, Conferences, and Exhibitions), where event organizers, suppliers, and attendees gather for academic, professional, cultural, or business purposes (Wee, Kamarulzaman, & Anas, 2022). Because a single venue must host this breadth of gatherings, its interior must accommodate a wide range of spatial and functional demands, placing the convention center's internal configuration, rather than its economic output, at the center of its usefulness.

MICE Industry

The sustained global growth of the MICE industry has driven continued investment in convention infrastructure (Allied Market Research, 2019), and with that growth have come rising expectations for the venues themselves. What increasingly distinguishes a competitive venue is less its economic scale than the quality and adaptability of its interior environment.

Multiple studies have noted that governments all across the globe have recognized the potential of the MICE industry, which has then led to investments in buildings for MICE venues (Lee & Fenich, 2016). However, as the MICE industry evolves through time, older facilities are now subjected to the pressure to modernize and adapt to meet the changing expectations of both the event organizers and attendees. Studies emphasized how important aligning the facilities with client demands is; this would include flexibility, cleanliness, technological integration, sustainability, and accessibility (An, Kim, & Hur, 2021).

The MICE industry can be correlated with event spaces in general, as it is essentially a space for various events. With this, a major consideration for the success of this industry would be the decision-making process of event organizers or planners, who act on behalf of their clients. These professionals consider various factors when it comes to selecting a venue, some of which would include the location, accessibility, and facilities (Lee & Lee, 2017). Additionally, adaptable room sizes, advanced audio-visual equipment, and user-friendly layouts play critical roles in ensuring long-term competitiveness in the MICE industry (Anas et al., 2020).

Modularity, Flexibility, and Adaptability

Modular design is a design strategy and common trend in convention center designs that involves either the combination or the separation of areas to cater to different kinds of events (Barr, 2023). Modular design also plays an important role when it comes to enhancing a convention center's flexibility and adaptability.

Flexibility and adaptability fall under the umbrella of modular design, wherein modular systems inherently support spatial reconfiguration, event customization, and functional upgrades without requiring substantial structural changes. Hashemian (2005) explains that modularity allows changes in one area to remain localized, thereby making a system easier to adapt. She further states that:

> "A product with a segmented architecture is generally easier to modify, hence has greater adaptability."

This segmentation can then enable a space to respond to varying functional demands actively, thereby reducing the need for rigid and fixed layouts.

Convention Centers are essentially multi-use spaces that cater to a diverse array of events, from corporate conferences and trade shows to cultural exhibits and community assemblies (Ike et al., 2023; Van der Wagen & White, 2017). To remain competitive and relevant in the evolving MICE industry, the venues should be designed with the future in mind. Hence, modular design becomes a strategic approach that facilitates the seamless transformation of spaces through demountable partitions,

adjustable seating, convertible stages, and technological integration (Ike et al., 2023; FXCollaborative, 2022).

In this context:

- Modularity provides the physical framework through which change is made possible.
- Flexibility enables the day-to-day reconfiguration of space to suit different activities.
- Adaptability ensures that the design can evolve, accommodating new technologies, user expectations, and environmental conditions.

With this in mind, modularity, flexibility, and adaptability are connected. Together, they allow convention centers to become resilient, sustainable, and user-centric environments that support a wide range of functions without compromising performance or identity (Hashemian, 2005; Ike et al., 2023). As emphasized in multiple studies, prioritizing flexibility through modular and adaptive design strategies will not only boost functionality but also promote ecological and economic growth, making these venues truly future-proof

Future-Proof Design

Future-Proof design refers to the strategic planning and creation of built environments that remain relevant, functional, and adaptable in the face of evolving needs and conditions. Designing with future-proofing in mind anticipates changes in technology, demographics, and environmental conditions (DesignHorizons, 2024). This approach ensures that spaces do not become obsolete but instead continue to serve users effectively over time.

A core principle of future-proofing is versatility and flexibility, which allows spaces to be reconfigured or repurposed without requiring significant structural changes. With the rapid development of society and technology, public spaces must be prepared to accommodate different functional requirements and usage methods at different times (Dong, Ibrahim, & Azahari, 2024). This might involve modular layouts, movable partitions, smart infrastructure, or adaptable lighting and climate systems.

By considering long-term trends, such as climate change, population growth, digital transformation, and shifts in work and lifestyle habits, future-proof design fosters sustainability and resilience. It minimizes the need for costly renovations, reduces environmental impact, and supports inclusivity and accessibility for a diverse range of users.

Ultimately, future-proofing embeds adaptability into the design itself, allowing spaces to remain functional and relevant as user needs, technologies, and environmental conditions change over time.

Adaptable & Flexible Design

Multiple studies have repeatedly highlighted two key design elements essential when designing adaptable and flexible spaces: technological integration, particularly in lighting and environmental control, and the use of movable furniture and partitions to transform spaces. Features like adjustable and dimmable lighting fixtures and smart lighting systems allow for the ambiance of a space to be tailored to suit different events. These technologies not only enhance visual comfort but also contribute to energy efficiency and user satisfaction by intelligently responding to the type of event or

occupancy levels (DesignHorizons, 2024; Dong, Ibrahim, & Azahari, 2024). Moreover, Pembegül (2009) states that

> “Interior design and technological equipment should meet the contemporary requirements of convention activities.”

Additionally, acoustics and spatial adaptability are just as essential. The incorporation of movable acoustic panels made from sustainable materials like recycled felt or cork supports effective sound management in multipurpose spaces, ensuring the functionality of the space across various events (DesignHorizons, 2024). Moreover, adjustable furniture and fixtures, such as foldable tables, stackable chairs, and movable partitions or sliding walls, provide spatial flexibility by allowing users to reconfigure the space quickly and efficiently. These elements can support transitions between open and private setups, catering to a wide range of user needs, from professional meetings to social gatherings (Dong, Ibrahim, & Azahari, 2024).

Acoustic performance is equally central to a reconfigurable event hall, and recent studies treat it as a spatial and architectural design problem rather than a purely engineering one. Cairoli (2020) documents a convention-center auditorium reconfigured with movable curtains, pivoting wood panels, and adjustable ceiling plates to shift between speech-oriented congress and music-oriented concert settings, while Habibi (2025) shows how curved panels and an adjustable ceiling can alter room volume to suit dual speech-and-music functions. Passive variable systems, operable panels and retractable elements that change a room's absorption, have been shown to meaningfully affect reverberation and speech intelligibility in multipurpose halls At the material scale, hybrid absorptive–diffusive wooden panels with interchangeable overlays allow a

surface's absorption class to be adjusted using sustainable, FSC-certified timber (Chojnacki et al., 2023, 2024), aligning acoustic control with the study's future-proof and sustainable-material goals.

Anthropometric and Ergonomic Standards

To develop a highly adaptable, culturally responsive, and operationally efficient interior design framework for the Calbayog City Convention Center's 840 sqm. Main Hall, the spatial layouts must be grounded in human scale, physiological capabilities, and regulatory compliance. These standards dictate the limits of physical comfort, universal accessibility, and egress safety within the interior envelope.

In public assembly interiors, human body dimensions govern the design of furniture, fixtures, and equipment (FF&E), along with the spatial clearances surrounding them. Anthropometry is broadly categorized into two fields of study, both of which are critical to the execution of flexible interior layouts:

- Static (Structural) Anthropometry: The measurement of the human body at rest in standardized, stationary positions. These metrics dictate the dimensions of individual seating modules, table heights, and structural sightlines.
- Dynamic (Functional) Anthropometry: The measurement of the body in motion, describing the spatial envelopes required for physical activities, reach zones, and movement patterns. These metrics dictate the clearance corridors between seating rows, the walking paths of service staff, and the physical force required to rearrange modular interior systems.

The 840 sqm Main Hall of the Calbayog City Convention Center must morph seamlessly between diverse layout configurations. To make these reconfigurations

viable, the physical layout templates must integrate the anthropometric clearances established by Panero and Zelnik (1979) and Neufert (2012).

A. Theater (Plenary) Layout Typology

- Seat Pitch (Row-to-Row Distance): The minimum distance from backrest to backrest must be calibrated to ensure both seated thigh clearance and emergency passage. Under standard conditions (Neufert, 2012), a seat pitch of 900 mm (35.4 in) is required. This provides a clear path of at least 305 mm (12 in) between the front edge of one seat cushion (when down) and the back of the seat in front.
- Lateral Seat Allocation: To comfortably accommodate 95th percentile male shoulder breadth and hip width, individual seat modules must have a minimum width of 500 mm (19.7 in), with 550 mm (21.6 in) preferred for executive event formats.
- Sightline Clearance (C-Value): The vertical sightline parameter must allow an observer to view the performance over the head of the person sitting directly in front. The standard vertical offset (C-value) between rows of seating is calculated at 120 mm (4.7 in), or 60 mm (2.4 in) with staggered seating layouts.

B. Classroom (Lecture) Layout Typology

- Desk-to-Chair Workspace: To accommodate seated reach and forward-leaning actions, the desk writing surface must be positioned 740 mm to 760 mm (29 in to 30 in) above the floor (Neufert, 2012).
- User Seat Clearance: The clearance depth between the front edge of the writing desk and the backrest of the seat behind it must be 450 mm to 500 mm (17.7 in to 19.7 in) to accommodate the abdominal depth and upper torso movement of a 95th-percentile occupant.
- Service Aisle Clearance: The corridor behind the occupied chair to the edge of the next desk must be at least 400 mm to 450 mm (15.7 in to 17.7 in). This dynamic clearance allows a service planner or late attendee to traverse the aisle sideways without disrupting seated occupants.

C. Banquet (Social/Dining) Layout Typology

- Table Clearance Circles: The centerpiece of the banquet typology is the standard circular table. For a 10-person round table (1500 mm to 1800 mm / 60 in to 72 in), a total circular spatial footprint of 3200 mm to 3500 mm (126 in to 138 in) must be maintained.
- Table-to-Table Service Lanes: The linear passage distance between adjacent circular tables must be a minimum of 1620 mm to 1800 mm (63.8 in to 70.8 in). This spacing accounts for back-to-back seated occupants (each requiring 450 mm of depth from the table edge) plus a central 720 mm to 900 mm walking

zone. This central zone accommodates the dynamic body width of servers carrying trays or planners navigating the room.

D. Exhibition Layout Typology

- Pedestrian Grid Corridors: To allow natural, non-congested foot traffic within the 840 sqm Main Hall during commercial setups, the circulation aisles between booth modules must be designed in multiples of the human shoulder-width profile.
- Pedestrian Clearance Multiples:
    - Single file passage: Minimum clear path of 750 mm to 900mm (29.5 in to 35.4 in) based on maximum body breadth.
    - Two-person bi-directional passing: Minimum path of 1500 mm to 800 mm (59.0 in to 70.8 in).
    - Three-person crowd capacity corridor: Minimum path of 2400 mm (94.5 in). This ensures that visitors can stop to engage with an exhibition booth without creating bottlenecks for passing guests.

To ensure the Calbayog City Convention Center is inclusive, the interior design overlay must prioritize universal design principles, making the space fully usable for individuals with physical, sensory, or cognitive challenges. Spatial modules must be designed to comply with local accessibility laws, specifically Batas Pambansa Bilang 344 (BP 344), and international benchmarks like the ADA Standards for Accessible Design (2010):

- Wheelchair Footprint and Clear Path: Under BP 344, a standard manual wheelchair has a clear structural footprint of 700 mm x 1200 mm (27.5 in x 47.2 in). To allow unhindered movement, walkways, primary ramps, and internal corridors must maintain a minimum clear width of 1200 mm (47.2 in).
- Wheelchair Turning Space: To execute a 360° direction change, a wheelchair user requires a turning circle with a minimum diameter of 1500 mm (59.0 in). This space must be built into dead ends, cross-aisles, and modular furniture clusters.
- Accessible Assembly Seating Dispersal: BP 344 and the ADA require that dedicated wheelchair parking stalls be seamlessly integrated into the general seating arrangement. These stalls must:
    - Have a minimum clear footprint of 760 mm x 1220 mm (30 in x 48 in) for rear or front-access entry.
    - Be located alongside a general circulation aisle to facilitate easy emergency egress.
    - Provide companion seating immediately adjacent to the wheelchair space.

Ceiling Design and Rigging Systems in Exhibition and Event Spaces

The ceiling design of interior spaces in large-scale venues like convention centers offers both aesthetic and functional opportunities. In exhibition and event settings, ceiling elements serve not only to conceal unsightly infrastructure like pipes and ductwork but also to elevate the visual and spatial experience (Syma, 2023). Modular ceiling configurations, such as raised ceilings, suspended panels, or ceiling sails, can enhance the branding, spatial legibility, and atmospheric quality of a venue while simultaneously allowing for fire safety and other building regulations to be met. These elements can be customized with mesh materials or branded banners, turning overhead space into an active zone for user engagement and visual communication.

Motorized truss systems add another layer of spatial adaptability. Institutional examples such as those at Hills Adventist College and UNSW's Webster 334 show how overhead rigging can be raised or lowered to reconfigure a space and ease maintenance (Cat, 2023; Phantos, 2025). From an interior standpoint, the value of such systems is spatial and aesthetic rather than mechanical; they keep visual clutter concealed within the ceiling plane, allowing lighting and drapery to be repositioned without disrupting the room below (Phantos, 2025).

More broadly, rigging and truss systems support flexible spatial design by carrying audiovisual elements—lights, speakers, and projection screens—overhead, freeing the floor for more fluid, open configurations across a variety of event types (AV Alliance, 2024). Because the present study treats the overhead framework as a spatial and aesthetic interface rather than an engineered assembly, the relevant lesson from

these systems is how they enable rapid reconfiguration of a venue's ambiance and lighting, not their load-rated hardware specifications.

The implementation of these systems supports the vision of future-proofed, modular interior design. By enabling quick reconfiguration, safer installations, and greater spatial responsiveness, these ceiling and rigging solutions play a vital role in the adaptability and sustainability of modern convention centers.

Events in a Convention Center

Convention centers are widely recognized as multi-use spaces that cater to a broad range of events like exhibitions, meetings, conferences, conventions, trade shows, cultural exhibitions, concerts, and industry expos, among others (Sharma, 2025; Fulgar, 2024; Cailles, 2024). This event diversity is precisely what makes interior configurability essential: each format below imposes different spatial, seating, and circulation demands on the same hall. Building on this, Jain (2015) identifies several of the most commonly held event types:

- **Professional Conventions**: Focused on issues of concern to a particular profession, these conventions aim to foster advancement and development within the field.



Figure 2-1. Image of a Professional Convention (Image Source: Trade Show News Network)

- **Fan Conventions**: Centered around pop culture, these events often feature celebrity guests, displays, performances, and merchandise related to popular media.



Figure 2-2. Image of a Fan Convention (Image Source: Study Breaks Magazine)

- **Seminars**: Organized gatherings designed to educate participants on a particular topic or skill, typically featuring expert speakers who share their knowledge and expertise.

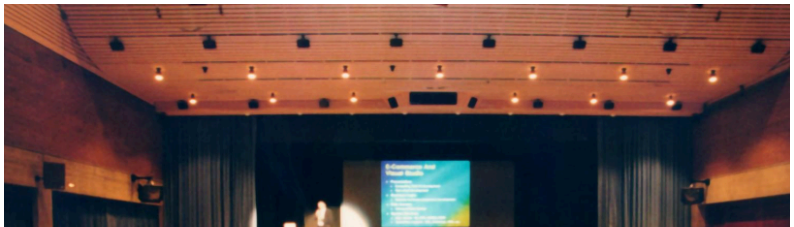


Figure 2-3. Image of a Seminar (Image Source: Globalinx)

- **Social Events**: Large-scale celebrations marking significant life milestones, such as anniversaries, weddings, and birthdays.



Figure 2-4. Image of a Wedding in a Convention Center (Image Source: White Eagle Events)

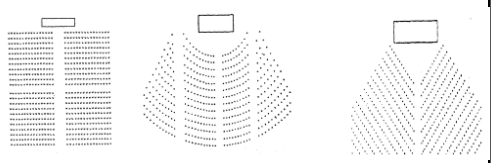
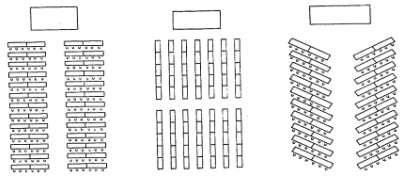
- **Trade Shows/Exhibitions**: Platforms where companies present their latest products and prototypes to journalists, industry professionals, and potential buyers, often highlighting innovations not yet released to the public.

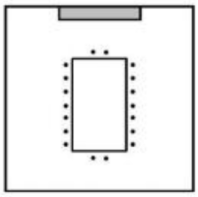
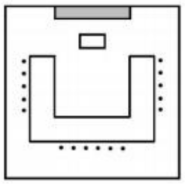
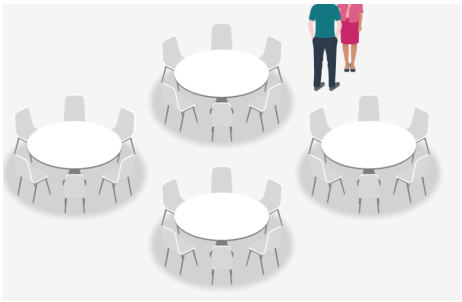


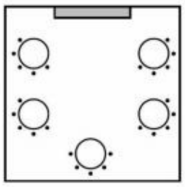
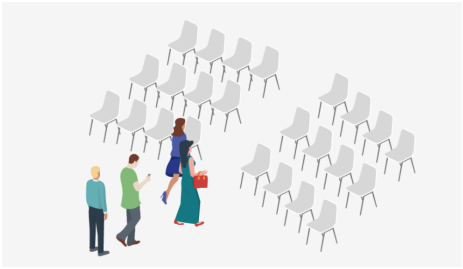
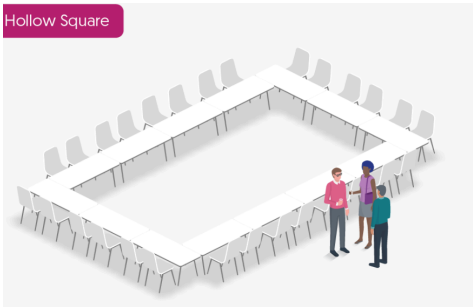
Figure 2-5. Image of a Trade Show (Image Source: Business World)

Seating Styles/ Layouts

According to Pembegül (2009), flexibility in the layout of meeting halls is essential to accommodate a variety of events. Across the literature, three studies, Pembegül (2009, drawing on Aymankuy, 2006), Jain (2015), and Waida (2021), describe a recurring set of seating configurations, each suited to a particular mode of interaction and event type. Consolidating these overlapping accounts yields the following core layouts:

Seating Layout	Best Suited For	Key Trade-off	Source(s)
Theater / Auditorium  Figure 2-6. Theater style, conventional- semi-circle shape- V-shaped (Pembegül, 2009; Aymankuy, 2006)	Large audiences; lectures, product launches, ceremonies	Maximizes capacity but limits interaction and sound control	Pembegül, 2009; Jain, 2015; Waida, 2021
Classroom  Figure 2-7. Classroom style, conventional- vertical- V-shaped (Pembegül, 2009; Aymankuy, 2006)	Note-taking; training, certifications, workshops	Supports learning but can feel cramped if poorly spaced	Pembegül, 2009; Waida, 2021
Conference / Boardroom  Figure 2-8. Conference style, (Pembegül, 2009; Aymankuy, 2006)	Interactive discussion; debates, negotiations, strategy meetings	Encourages dialogue but seats limited numbers	Pembegül, 2009; Jain, 2015; Waida, 2021

Seating Layout	Best Suited For	Key Trade-off	Source(s)
 Figure 2-9. Boardroom style layout (Jain, 2015)			
U-Shaped  Figure 2-10. U-shaped layout (Jain, 2015)	Presentation plus participant interaction; team meetings, workshops	Balances focus and discussion but weak beyond ~30 people	Jain, 2015; Waida, 2021
Banquet  Figure 2-11. Banquet style layout (Waida, 2021)	Socialization and dining; weddings, awards, charity events	Fosters networking but limits cross-table interaction	Waida, 2021
Cabaret	Relationship-building with a retained focal point; relaxed	Engaging but does not maximize floor space	Jain, 2015; Waida, 2021

Seating Layout	Best Suited For	Key Trade-off	Source(s)
 Figure 2-12. Cabaret style layout (Jain, 2015)	shows, entertainment		
Chevron  Figure 2-13. Chevron style layout (Waida, 2021)	Speaker focus with small-group discussion; training, interactive presentations	Space-efficient but risks sightline issues if unstaggered	Waida, 2021
Hollow Square  Figure 2-14. Hollow square layout (Waida, 2021)	Collaboration where all participants see one another; group meetings, dining	No single focal point; weak beyond ~30 people	Waida, 2021

Cultural Identity

Convention centers are not just used for events, they can also become a platform for expressing cultural identity and fostering community pride. Several case studies highlight how architecture and programming can integrate and celebrate local culture.

In the Philippines, convention centers act as hubs of cultural and economic connectivity. Fulgar (2024) emphasizes that these venues are not just for commerce and trade but are pivotal spaces where Filipino culture and identity are showcased. The Philippine International Convention Center (PICC), designed by National Artist Leandro Locsin, remains an enduring symbol of the country's global aspirations, cultural pride, and architectural heritage. As gathering spaces for diverse sectors, from policymakers to artists, Philippine convention centers promote knowledge exchange and community engagement, reinforcing local identity while supporting national development.



Figure 2-15. Photo of the Philippine International Convention Center (Image Source: PICC)

At the Albuquerque Convention Center in New Mexico, cultural celebration is embedded deeply into its operation and design. The center regularly hosts multicultural events like Diwali, the Gathering of Nations, Mariachi Spectacular, and Juneteenth celebrations, positioning itself as a key venue for cultural exchange. It also integrates public art by New Mexican artists, with nearly 200 artworks displayed throughout the facility. These elements not only enrich visitor experience but also transform the center into a living museum of local and indigenous traditions, reinforcing a strong sense of place and belonging within the downtown Albuquerque community (ASM Global, 2024).

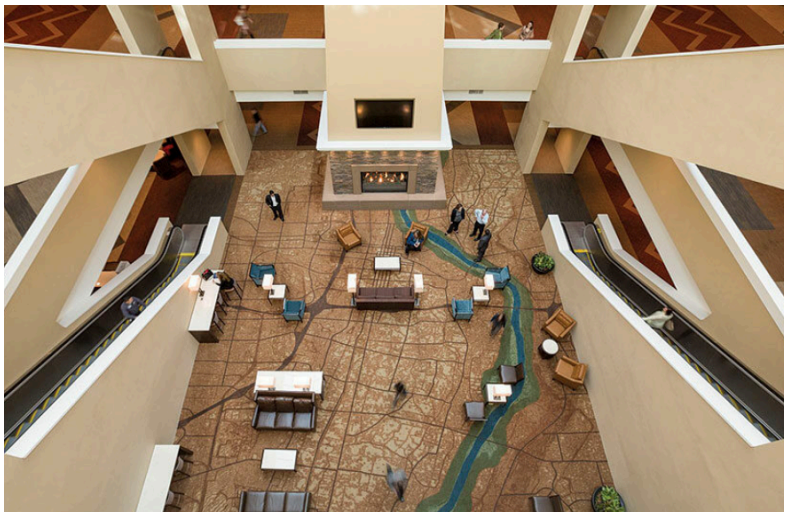


Figure 2-16. Photo of the Albuquerque Convention Center (Photo by: Robert Reck)

Heatherwick Studio's redesign of Seoul's Coex Convention Centre places a strong emphasis on cultural resonance and placemaking. Their winning proposal reimagines the venue as a "Cabinet of Curiosities," moving away from its generic glass facade to a modular, inviting structure that promotes discovery and public interaction (Fakharany, 2025). By blending event spaces with public gardens, viewing decks, and cultural exhibits, the new Coex aims to serve not just event participants but the wider

community. The design strategy intentionally strengthens ties with the surrounding park and neighborhood, ensuring that Coex becomes an integral, culturally relevant urban landmark even outside of major events.



Figure 2-17. Photo of Coex Convention Center (Image Source: Arch Daily)

Taken together, these precedents show culturally-responsive design operating at national (PICC), community and indigenous (Albuquerque), and placemaking (Coex) scales. For the Calbayog City Convention Center, the equivalent move is to translate this principle to the Samar region, embedding local heritage directly into the interior surfaces, which is the focus of the following section.

Culturally Responsive Design

Culturally responsive design is an approach that integrates the values, traditions, and spatial practices of a particular culture into the built environment, ensuring that architecture resonates with its users' identities and ways of life. This approach moves beyond aesthetics to address the deeper social and cultural meanings embedded in

space, contributing to user well-being and social sustainability (Krstić, Filipović, & Ristić Trajković, 2024).

Kockler (1989), in her study on culturally responsive architecture for the Latino community in Roxbury, emphasized that design should reinforce core cultural attributes while adapting them to contemporary contexts. The failure to do so can lead to environments that feel alienating and disconnected from their users. Instead, she argued for spatial strategies that promote cultural continuity and empower communities in the design process.

The integration of these principles into interior design is particularly crucial, as interiors constitute the most immediate and intimate environments where individuals interact, live, and express identity. Kockler (1989) demonstrated that cultural values can be reflected in the use of specific materials, art forms like murals or mosaics, and spatial configurations that resonate with collective memory and heritage. Interior design thus becomes a medium through which occupants can recognize themselves in the built form.

While Kockler's study centers on domestic settings, its underlying principles transfer to civic and communal interiors, and it is those principles, rather than the specific residential forms, that this study draws on. Kockler (1989) illustrated the role of spatial organization in reinforcing cultural values through the prioritization of extended family structures in interior layouts, the use of open courtyards, and the delineation of space through a hierarchy from public to private zones. These configurations support cultural practices such as shared meals, intergenerational cohabitation, and communal

interaction, which are practices that are similarly observed in many Filipino households and community structures.

The relevance of culturally responsive design extends beyond residential typologies to public buildings and community spaces. Interior layouts in such spaces must accommodate diverse functions (legal aid, education, recreation, and commerce) while remaining culturally resonant and emotionally engaging (Kockler, 1989). Similarly, Krstić et al. (2024) emphasized that emotionally responsive, culturally sensitive environments can enhance occupants' sense of belonging and promote well-being. In modular or adaptive architectural typologies, this means creating flexible spatial arrangements that still reflect cultural narratives, symbols, and rhythms.

Samar Weaving Heritage and Its Translation into Interior Surfaces

Within the Philippine context, the culturally-responsive principles above find a concrete anchor in the weaving heritage of Samar. Basey, Samar, is widely recognized as a center of banig (mat) weaving, a craft that uses locally grown tikog grass and functions simultaneously as a livelihood and as a marker of community identity passed down across generations (Garado et al., 2026; Del Monte et al., 2022). This living tradition supplies both a material vocabulary (woven fibers and mat textures) and a repertoire of geometric patterns that can inform interior surface treatments without dictating spatial layout.

Translating such heritage into the built environment is an established design method rather than a purely decorative gesture. Wagas (2016) demonstrates a replicable process for abstracting an indigenous weave into a reproducible pattern

language, showing how a traditional motif can be systematically converted into a surface-design data set. At a broader scale, Baradas (1968) documents how a single regional motif, the Maranao okir, recurs coherently across weaving, carving, and architecture while carrying consistent symbolic meaning, an enduring precedent for embedding a local motif into architectural and interior surfaces. Del Monte (2021) situates this within contemporary Filipino practice, examining how designers negotiate imported design norms against the expression of local cultural identity in the built environment.

For the Calbayog City Convention Center, these sources support a strategy in which Samar's banig and tikog weaving motifs are abstracted and applied to permanent and semi-permanent interior surfaces, wall panels, floor patterning, and ceiling finishes, so that the venue communicates a distinct sense of place while its layout remains fully reconfigurable. Documentation from the National Commission for Culture and the Arts on regional mat-making traditions further grounds this material selection (NCCA, 2004); however, a verifiable typology of specifically Waray/Eastern Visayas weaving motifs remains thin in the published literature and may warrant primary documentation (e.g., interviews with Basey weavers or the NCCA intangible-heritage inventory).